

AI OPS Assignment 3

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1 Ray vs Spark Performance in the Experiment

The Performance Graphs are as below. I ran identical dataset cleaning task functions on both the ray and spark workers and they performed quite similar in terms of time taken. The dataset cleaning task was to remove duplicates and null values and a table join and compute a new column which was average speed and involved doing some function calculation on the already existing columns in the dataset. This experiment made use of the yellow taxi dataset and the locations data for the table join. All the screenshots as well as the code used have been included in the code repository. The graphs below are the plotted versions of the numbers I obtained by running the experiments.

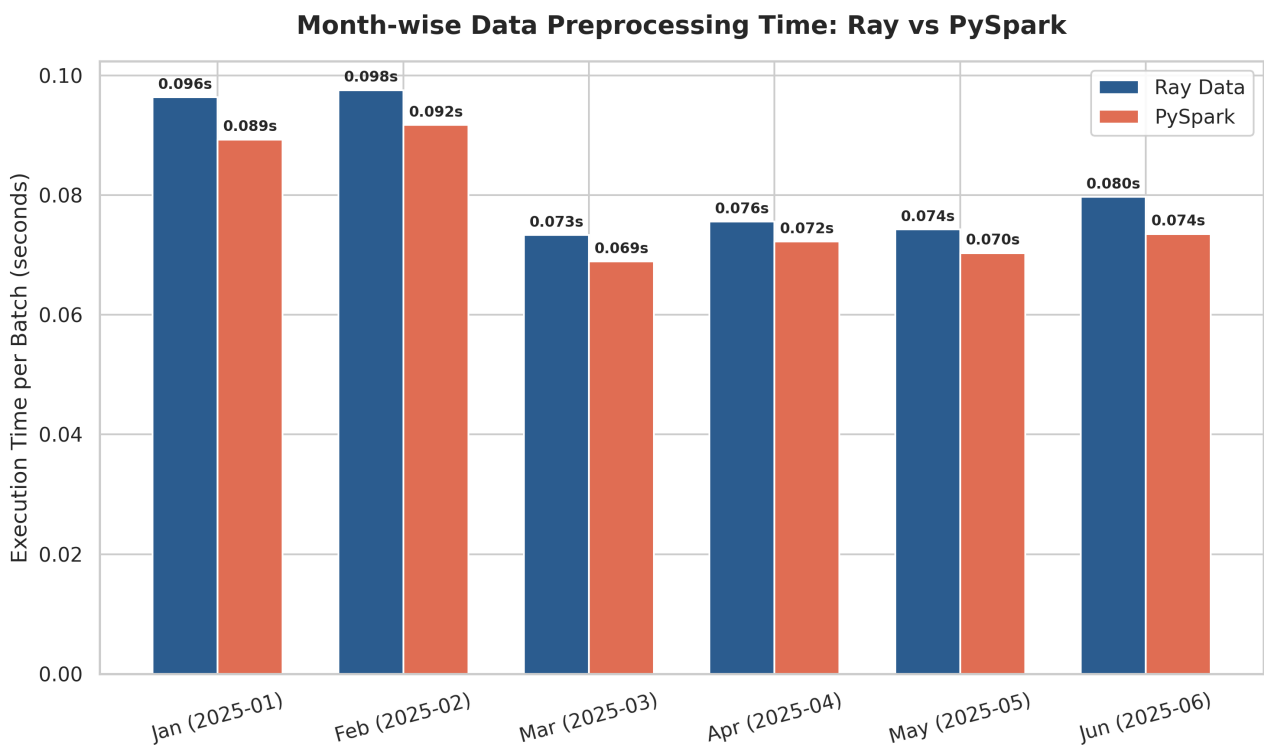


Figure 1: UDF Function Statistics Per Month

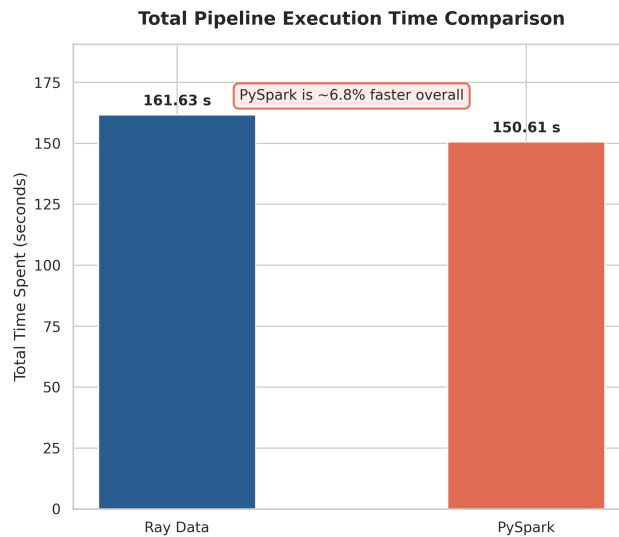


Figure 2: Time Taken by Ray and Spark

I think that while Ray is more modern and Spark is a relatively older framework, it could have been faster for this use case because it did not need to use its own data format like Ray had to use the ray data format but since the difference in time is so small it is likely due to small optimizations in Spark which were more suited for this specific scenario.

2 BI First vs AI First

BI First refers a type of structure which prioritizes SQL Queries and table transformations and does not use AI. For this structure I think Spark would perform better as it is good at Data Parallelization of the tasks.

Meanwhile AI First refers to a type of structure where there is a heavy use of AI and different models and thus the Ray Framework which is modern and created for AI is likely to handle it better.